Balancing Chemical Equations

Counting atoms, balancing equations and writing equations from words

Name _____ Class _____ Date _____

STEPS FOR BALANCING

1. Write correct formulas. 2. Count atoms on each side.
3. Balance one element at a time, leaving H, O and free elements until last.
4. Re-count everything. 5. Use the smallest whole numbers.

Change coefficients only. Never change subscripts.

A. Understanding

1. Explain why a chemical equation must be balanced.

2. What is the difference between the 2 in $2\,H_2O$ and the 2 in H_2O? Why can you change only one of them when balancing?

B. Counting atoms

3. Count the atoms of each element in (a) $4\,NH_3$ (b) $3\,Mg(OH)_2$ (c) $2\,Fe_2(SO_4)_3$

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C. Balancing

4. Balance: (a) $N_2 + H_2 \longrightarrow NH_3$ (b) $Mg + O_2 \longrightarrow MgO$ (c) $H_2O_2 \longrightarrow H_2O + O_2$ (d) $Fe + Cl_2 \longrightarrow FeCl_3$

5. Balance: (a) $Al + O_2 \longrightarrow Al_2O_3$ (b) $KClO_3 \longrightarrow KCl + O_2$ (c) $Zn + HCl \longrightarrow ZnCl_2 + H_2$

6. Balance these combustion reactions: (a) $C_2H_4 + O_2 \longrightarrow CO_2 + H_2O$ (b) $C_2H_5OH + O_2 \longrightarrow CO_2 + H_2O$ (c) $C_4H_{10} + O_2 \longrightarrow CO_2 + H_2O$

7. Balance, treating polyatomic ions as units: (a) $AgNO_3 + CaCl_2 \longrightarrow AgCl + Ca(NO_3)_2$ (b) $H_2SO_4 + NaOH \longrightarrow Na_2SO_4 + H_2O$ (c) $Al_2(SO_4)_3 + Ca(OH)_2 \longrightarrow Al(OH)_3 + CaSO_4$

D. From words to equations

8. Sodium reacts with water to form sodium hydroxide and hydrogen gas. Write the balanced equation.

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E. Challenge

9. (a) Is $\mathrm{CaCO_3 + HCl \longrightarrow CaCl_2 + H_2O + CO_2}$ balanced? If not, balance it. (b) Balance the equation for photosynthesis: $\mathrm{CO_2 + H_2O \longrightarrow C_6H_{12}O_6 + O_2}$

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